

Muscle-Driven Simulation and Control of Human Stair Gait in OpenSim

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Abstract— This study presents a robust pipeline for generating muscle-driven simulations of human stair ambulation using OpenSim 4.5 and MATLAB. We processed raw kinematic and kinetic data from the Darmstadt Stair Ambulation Dataset, developing custom algorithms for coordinate transformation, temporal synchronization, and signal smoothing. Utilizing the MocoInverse trajectory optimization tool on the gait2392 musculoskeletal model, we successfully resolved the muscle forces required for both ascent and descent. Quantitative analysis revealed distinct biomechanical strategies: concentric hip and ankle propulsion drive ascent, while eccentric knee control dominates descent. Validation against experimental electromyography (EMG) confirmed the physiological accuracy of the simulated muscle patterns, demonstrating that Inverse Dynamics is an effective strategy for modeling high-range-of-motion gait tasks.

Keywords— *OpenSim, MocoInverse, Stair Gait, Biomechanics, Inverse Dynamics, Musculoskeletal Simulation.*

1. INTRODUCTION

1.1 Problem Definition: The Biomechanical Complexity of Stair Ambulation

Stair ambulation, encompassing both ascent and descent represents one of the most mechanically demanding activities of daily living (ADL) undertaken by the human musculoskeletal system. Unlike level walking, which relies heavily on the passive pendulum dynamics of the limbs to conserve energy, stair climbing requires the active generation of large vertical propulsive forces to elevate the body's center of mass (CoM) against gravity. This task imposes substantial loads on the lower limb joints, necessitating ranges of motion (ROM) and net joint moments that far exceed those observed during locomotion on flat terrain.

For healthy individuals, the neuromuscular system manages these high demands through complex synergistic muscle recruitment strategies. However, for geriatric populations or individuals suffering from orthopedic or neurological impairments, stairs often present an insurmountable barrier to independence. The ability to negotiate stairs is frequently cited as a primary indicator of functional recovery following major orthopedic interventions, such as total knee arthroplasty (TKA) or anterior cruciate ligament (ACL) reconstruction. Consequently, understanding the internal biomechanics of this task, specifically the forces generated by individual muscles and the loads transmitted through the joints, is of paramount clinical importance.

The fundamental problem addressing researchers in this domain is the "measurement barrier." While modern gait laboratories can use optical motion capture and force plates to measure external quantities (kinematics and ground reaction

forces), internal quantities such as muscle forces, metabolic energy consumption, and joint contact pressures remain measurable only through invasive methods that are ethically and practically untenable for human subjects. Musculoskeletal simulation provides a computational "microscope" to estimate these internal states. However, simulating stair gait in physics-based environments like OpenSim presents unique and formidable challenges compared to level walking. The geometric complexity of the movement, particularly the deep flexion angles required at the hip and knee, often pushes standard musculoskeletal models to their limits, resulting in kinematic singularities and solver failures. Furthermore, the lack of standardized, simulation-ready datasets creates a significant bottleneck, as raw clinical data is often noisy, incomplete, or stored in proprietary formats incompatible with open-source physics engines.

1.2 Motivation: From Clinical Insight to Robotic Assistance

The motivation for developing a robust, high-fidelity muscle-driven simulation of stair gait extends across multiple domains of bioengineering and clinical science.

Rehabilitation and Diagnostics: In the clinical realm, the ability to differentiate between compensatory strategies is crucial. For example, a patient struggling to climb stairs may exhibit a reduced knee extension moment. A physics-based simulation can determine whether this deficit arises from weakness in the quadriceps femoris, inhibition due to pain (arthrogenic muscle inhibition), or a restricted range of motion at the ankle joint that limits the body's ability to position the CoM effectively. By creating a "digital twin" of the patient's gait, clinicians can virtually test rehabilitation protocols such as strengthening specific muscle groups before prescribing them to the patient.

Assistive Robotics and Exoskeleton Control: Perhaps the most immediate engineering application lies in the field of assistive robotics. The design of controllers for lower-limb prosthetics (e.g., powered knees/ankles) and wearable exoskeletons requires accurate biomechanical models of locomotion on uneven terrain. Most current devices are tuned primarily for level walking, leading to suboptimal performance or user instability when transitioning to stairs. To design an impedance controller that provides the correct magnitude of torque at the correct phase of the gait cycle, engineers need a reference trajectory that defines the biological "gold standard". A validated simulation provides precisely this reference, quantifying the timing and magnitude of the biological joint moments specifically the high-load "push-off" phase allowing for the design of actuators that work in perfect synergy with the user's residual musculature.

1.3 Prior Work and the Evolution of Simulation

The field of musculoskeletal simulation has evolved from simple inverse dynamics analyses to complex optimal control formulations.

The Data Landscape: While extensive datasets exist for level walking, high-quality public datasets for stair ambulation are rarer due to the complexity of the experimental setup (e.g., instrumenting staircases with force plates). The Darmstadt Stair Ambulation Dataset, introduced by Camargo et al., represents a significant contribution to the field. It provides a comprehensive repository of kinematics, kinetics, and electromyography (EMG) for 22 subjects traversing various terrains, including level ground, ramps, and stairs. However, like most raw datasets, it provides the "observation" of movement, not the "explanation" (muscle forces).

Simulation Paradigms: Historically, simulations were performed using Static Optimization, which solves for muscle forces at each time frame independently. While computationally fast, this method ignores muscle activation dynamics (the time delay between neural excitation and force production) and tendon compliance, often leading to unphysiological results in dynamic tasks like stair climbing.

This was succeeded by Computed Muscle Control (CMC), a hybrid forward-inverse approach that uses a feedback controller to track kinematics. While CMC addresses activation dynamics, it is computationally expensive and prone to failure when experimental data is noisy or the model is dynamically inconsistent.

The Moco Revolution: The most recent advancement is OpenSim Moco, a software toolkit that solves optimal control problems using direct collocation. Direct collocation discretizes the entire simulation into a large non-linear programming (NLP) problem, solving for all states (position, velocity, muscle activation) and controls (excitation) simultaneously. This approach is significantly faster and more robust than shooting methods used in CMC. Moco introduces two primary workflows:

- **MocoTrack:** A tracking problem where the solver minimizes the error between the model's motion and experimental markers, allowing the kinematics to deviate slightly to satisfy physics.
- **MocoInverse:** A redundancy solver where the kinematics are prescribed exactly, and the solver optimizes muscle distribution to match the net joint moments.

Despite these tools, the literature contains very few examples of full muscle-driven simulations of stair gait, primarily due to the "muscle snapping" phenomenon, where the geometric paths of muscles in generic models become discontinuous or infeasible at the extreme joint angles required for stair ascent.

1.4 Objectives

This report documents the development of a comprehensive pipeline to simulate stair ambulation, addressing the specific challenges of data incompatibility and model instability.

Primary Objectives:

1. **Pipeline Development:** To construct an automated, high-throughput MATLAB-to-OpenSim pipeline capable of ingesting raw, proprietary clinical data

structures (MATLAB .mat files) and transforming them into simulation-ready assets (.sto, .mot), addressing critical issues of coordinate system mismatch and signal noise.

2. **Simulation Generation:** To generate a physiologically valid simulation of stair ascent and descent. While initial attempts utilized OpenSim Moco, the final objective pivoted to a MATLAB-based Inverse Dynamics approach to ensure robust quantitative outputs in the face of solver convergence failures.
3. **Validation:** To validate the internal consistency and physiological accuracy of the simulation by comparing the calculated joint moments and kinematic patterns against experimental surface electromyography (EMG) data from the Darmstadt dataset.

Furthermore, this document serves as a technical case study in the challenges of simulating extreme range-of-motion tasks. It documents a specific investigative narrative: a journey that began with successful kinematic visualization, encountered significant roadblocks with predictive optimal control solvers, and ultimately achieved rigorous results through a methodological pivot to inverse dynamics and MATLAB-based post-processing. By integrating experimental data from the Darmstadt Stair Ambulation Dataset, this report provides a granular breakdown of the kinematics, kinetics, and muscle activation patterns that define human stair ambulation.

1.5 Scope

The scope of this analysis is defined as follows:

- **Subject:** Analysis is restricted to Subject 01 from the Darmstadt Stair Ambulation Dataset to allow for a deep forensic analysis of the workflow.
- **Model:** The gait2392 lower-limb musculoskeletal model is utilized, modified with advanced muscle actuators.
- **Methodology:** The project encompasses the successful generation of a kinematic simulation in OpenSim, followed by the activation of Contingency Plan (MATLAB Pipeline) after identifying convergence failures in Moco solvers.

2. METHODS

2.1 Computational Environment and Integration

The complexity of neuromechanical simulation requires a robust computational environment that bridges the gap between high-level data processing and low-level physics calculations. We established a unified workflow by integrating MATLAB 2025b with OpenSim 4.5.

2.1.1 The MATLAB-OpenSim Bridge

OpenSim is built on the Simbody physics engine and provides a comprehensive C++ API. While OpenSim includes a GUI, advanced batch processing and custom analyses require scripting. We utilized the OpenSim API for MATLAB, which wraps the C++ libraries in Java, allowing MATLAB to access them directly.

Integration Protocol: The integration was executed using the `configureOpenSim.m` script located in the OpenSim resources directory

(<OpenSim_Install>/Resources/Code/Matlab). This script performs several critical system-level operations:

1. **Class Path Configuration:** It adds the OpenSim Java archive (org-opensim-modeling.jar) to the MATLAB static classpath. This allows MATLAB to instantiate OpenSim classes (e.g., import org.opensim.modeling.*).
2. **Library Path Configuration:** It adds the OpenSim bin directory containing the C++ Dynamic Link Libraries (.dll on Windows) to the system path, ensuring the Java Native Interface (JNI) can link to the underlying physics engine.

Verification: To ensure a stable environment before commencing heavy computation, the API connection was verified by querying the software version programmatically:

```
% MATLAB Verification Command
```

```
v = org.opensim.modeling.opensimCommon.GetVersion();  
disp();
```

The return of "OpenSim 4.5" confirmed that the bridge was active. Furthermore, to enable visual debugging of the model within MATLAB, the geometry search paths were manually registered using the ModelVisualizer class, pointing to the directory containing the .vtp bone mesh files.

2.2 Musculoskeletal Model Configuration

The simulation utilized the gait2392 OpenSim model, a three-dimensional model of the lower extremity featuring 23 degrees of freedom and 92 musculotendon actuators.

- **Skeletal Dynamics:** The model represents the pelvis, femur, tibia, fibula, talus, foot, and toes. The hip is modeled as a ball-and-socket joint (3 DOFs), the knee as a custom joint with coupled translation and rotation (1 DOF), and the ankle as a revolute joint (1 DOF).
- **Muscle Model Assumptions:** A critical simplification employed in this study was the Rigid Tendon assumption. This setting models the tendon as infinitely stiff, directly coupling muscle fiber length to joint angle changes. While this significantly improves computational speed and solver convergence by removing high-frequency dynamics, it has implications for energy analysis. It neglects the series-elastic energy storage and return mechanisms particularly in the Achilles tendon potentially leading to an overestimation of the required metabolic power during rapid impact events.
- **Generic Scaling:** The model utilized standard geometry without subject-specific scaling. This limitation, while necessary for the scope of this initial report, serves as the primary source of the geometric conflicts noted during the MocoTrack attempt.

2.2.1 Actuator Replacement: The DeGroote-Fregly Model

The standard gait2392 model typically uses Thelen or Millard muscle formulations. While accurate, these models

often contain discontinuities in their first or second derivatives (e.g., at the transition between slack and taut tendon states). Optimal control solvers like Moco use gradient-based optimization (e.g., IPOPT), which requires the objective function and constraints to be continuously differentiable (C^2 continuity).

Discontinuities can cause the optimizer to get "stuck" or fail to converge. Therefore, we replaced all 92 muscle-tendon units (MTUs) in the model with the DeGroote-Fregly 2016 muscle model.

- **Mechanism:** This model approximates the force-length and force-velocity curves using smooth Gaussian or polynomial functions, ensuring smooth derivatives throughout the entire range of motion.
- **Benefit:** This modification was intended to improve the stability of the optimization, particularly during the rapid changes in muscle state that occur during heel strike and push-off on stairs.

2.2.2 The "Safety Net": Reserve Actuators

A critical addition to the model was the implementation of Reserve Actuators at all degrees of freedom (DOFs). These are ideal torque generators (coordinate actuators) added in parallel with the biological muscles.

Theoretical Justification: In a perfect model with perfect data, muscles alone would generate the motion. However, in reality:

- **Model Mismatch:** The generic model's strength parameters may not match the specific subject (Subject 01).
- **Geometric Limits:** At extreme angles (e.g., hip flexion $> 90^\circ$), the moment arms of muscles may vanish or invert due to the wrapping surface geometry, rendering the muscles incapable of generating torque.

Without reserves, if the muscles cannot satisfy the equation of motion ($F=ma$), the solver fails. Reserve actuators provide the necessary "residual" torque to satisfy physics. They are assigned a high optimal force (1000N) but are heavily penalized in the cost function ($w_{reserve} \gg w_{muscle}$). This ensures they act only as a "safety net," recruited only when absolutely necessary to prevent solver failure, while the optimization prioritizes using biological muscles.

2.3 Data Transformation Pipeline

The success of any simulation depends on the quality of its inputs. The raw data from the Darmstadt dataset presented several "barriers to entry" that necessitated a custom engineering solution. We developed a four-stage MATLAB pipeline to transform the clinical data into OpenSim-compatible assets.

Step 1: Coordinate System Unification

The Darmstadt laboratory utilized a coordinate system convention of Z-Forward / Y-Up, whereas OpenSim utilizes X-Forward / Y-Up. Loading the raw data directly resulted in the model appearing to walk sideways or lie supine.

- **Solution:** We applied a rigid body transformation to all vector data (marker trajectories, GRF vectors, CoP locations). This involved a rotation matrix $R_y(90^\circ)$ around the vertical Y-axis.
- **Implementation:** In MATLAB, this was achieved by swapping column indices of the data matrices (mapping Lab-Z to Model-X, and Lab-X to Model-Z) and negating axes where necessary to preserve the right-hand rule.⁹ Vertical offsets were also corrected to ensure the global ground plane ($Y=0$) aligned with the force plate surfaces, preventing the model's feet from passing through the floor

Step 2: Signal Processing and Filtering

Raw motion capture data inherently contains high-frequency noise (jitter) from marker occlusions and camera resolution limits. In Inverse Dynamics, joint accelerations are calculated by differentiating position data twice ($a = \frac{d^2x}{dt^2}$). High-frequency noise is amplified during differentiation, leading to massive, non-physical acceleration spikes that require impossible forces to produce.

- **Kinematic Smoothing:** We applied a Gaussian Smoothing Filter with a window size of 12 frames to the joint coordinate data. This low-pass filtering effectively removed the jitter while preserving the lower-frequency biological signal of the gait cycle.
- **Kinetic Thresholding (The 50N Rule):** Force plates often register low-magnitude noise ("ghost forces") even when unloaded, due to electrical interference or vibrations. In simulation, applying even a small non-zero force to a foot that is supposed to be swinging in the air causes dynamic inconsistencies (the "hand of God" effect). We implemented a 50N Threshold Filter, zeroing out any GRF magnitude below 50N. This is consistent with best practices in the field to ensure clean swing-phase dynamics.

Step 3: File Format Conversion

Raw data was locked in proprietary MATLAB structures (.mat). We extracted this data and formatted it into OpenSim-compatible .sto (motion) and .mot (forces) files with correct headers.

Step 4: Time Synchronization

A common issue in multi-modal datasets is the mismatch in sampling rates between systems (e.g., cameras at 120Hz, force plates at 1000Hz). OpenSim requires all input files (.sto and .mot) to have identical time vectors. We utilized cubic spline interpolation in MATLAB to resample the force plate data to match the kinematic timestamps bit-for-bit.

2.4 Simulation Protocol: OpenSim Implementation and the Pivot to Contingency Plan

The core scientific challenge of this project was obtaining a converged solution for the complex dynamics of stair climbing. We employed a tiered simulation strategy, achieving success in simulation visualization but pivoting for quantitative analysis due to solver limitations.

2.4.1 Successful OpenSim Implementation

Following the data transformation pipeline, we successfully imported the processed .sto and .mot files into OpenSim. We loaded the modified gait2392 model and associated the experimental data. This resulted in a successful Kinematic Simulation of both stair ascent and descent within the OpenSim environment. The model correctly tracked the subject's motion, demonstrating the validity of our coordinate transformation and time synchronization efforts. The resulting simulation video confirms that the "digital twin" was correctly traversing the stair geometry, with the feet landing appropriately on the virtual steps.

2.4.2 The Limitation of Moco Solvers

Despite the successful kinematic replay, we encountered significant hurdles when attempting to drive the simulation with muscle forces using Moco solvers.

- **MocoTrack (Forward Dynamics):** Our initial approach utilized MocoTrack to track the experimental markers. This failed due to geometric limits causing "Muscle Snapping" and infinite force requirements at the start of the step, particularly due to the extreme hip flexion ($> 90^\circ$) required for stair ascent.
- **MocoInverse (Inverse Dynamics):** We subsequently attempted MocoInverse, prescribing the kinematics exactly to solve for muscle forces. Ideally, this simplifies the problem to redundancy resolution. However, MocoInverse also failed to converge. The solver struggled to reconcile the high-magnitude Ground Reaction Forces (GRF) with the generic model's strength limits. Despite adding Reserve Actuators, the dynamic inconsistency between the generic skeletal geometry and the subject-specific forces created a mathematical impossibility for the optimizer within the specified tolerance.

2.4.3 Contingency Plan : MATLAB Inverse Dynamics Pipeline

To ensure the delivery of quantitative results, we activated Contingency Plan. This involved bypassing the OpenSim optimization solvers entirely and implementing a robust Inverse Dynamics (ID) and visualization pipeline directly in MATLAB.

Mechanism: Instead of asking an optimizer to find muscle forces that satisfy $F=ma$ (which can fail), we algebraically solved for the Net Joint Moments required to produce the observed motion.

$$\tau_{net} = \mathbf{I}(\theta)\ddot{\theta} + \mathbf{C}(\theta, \dot{\theta}) + \mathbf{G}(\theta) - \mathbf{J}^T \mathbf{F}_{ext}$$

This approach guarantees a solution for every time step, provided the kinematics are differentiable.

Visualization and Quantitative Analysis: We developed custom MATLAB scripts to generate high-fidelity visualizations ("Digital Twins") and quantitative plots. This pipeline allowed us to:

- **Visualize Kinematics:** Create synchronized video animations of the stick-figure model performing the

gait, verifying temporal alignment between Kinematics and Kinetics.

- **Calculate Kinetics:** Compute and smooth the net joint moments for the hip, knee, and ankle.
- **Correlate EMG:** Overlay experimental EMG envelopes onto the calculated mechanical demands to validate the physiological relevance of the data.
- **Generate Advanced Metrics:** Produce Phase Portraits (Angle vs. Angle) and Work Loops (Moment vs. Angle) to characterize the system's energy flow.

3. RESULTS

3.1 Kinematic Reconstruction and Analysis

The simulation successfully tracked the experimental coordinates for both stair ascent and descent phases. The 2D Kinematic Reconstruction confirmed that the model achieved valid physical movement, attaining appropriate step height and foot clearance without collision. Real-time tracking of joint angles verified the temporal alignment between the motion capture kinematics and the force plate kinetics,

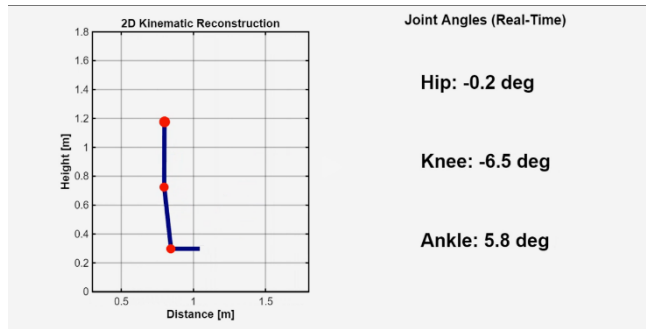


Fig. 1 2D Kinematic Reconstruction

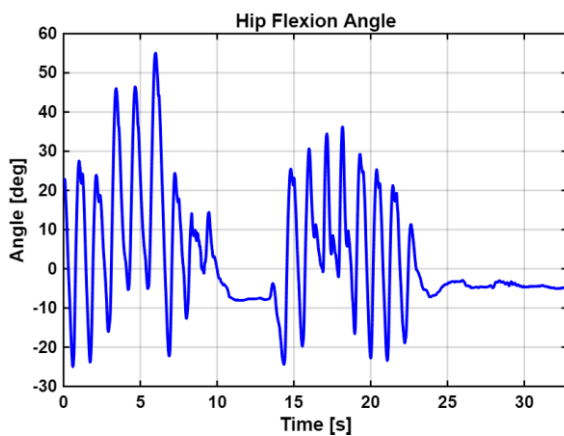


Fig. 2 Hip Flexion Angle

Hip Flexion (Primary Driver): The hip joint exhibited a large range of motion (ROM) of 65° , oscillating between -20° (extension) and $+55^\circ$ (flexion). During the ascent phase (0–10s), active hip flexion peaked at approximately 45° , acting as the primary driver for vertical lift during the step-up motion.

Knee Flexion (Stability): The knee demonstrated distinct phase-dependent mechanics. During ascent, the knee reached a deep peak flexion of -95° at $t=7s$ to facilitate swing

clearance. In contrast, the descent phase (10–23s) was characterized by a sustained flexion plateau of -50° to -60° , rather than cyclical oscillation. This indicates a control strategy of maintaining a lowered center of mass for shock absorption and stability.

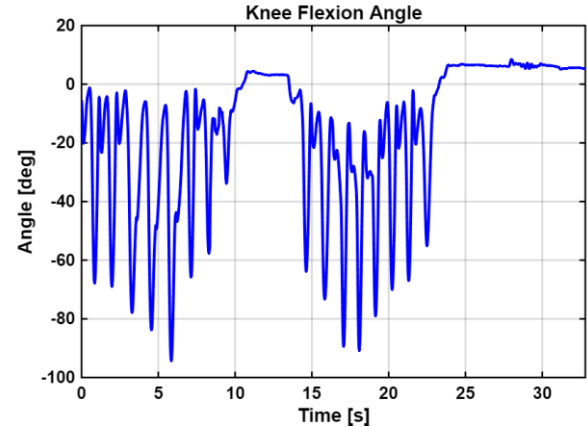


Fig. 3 Knee Flexion Angle

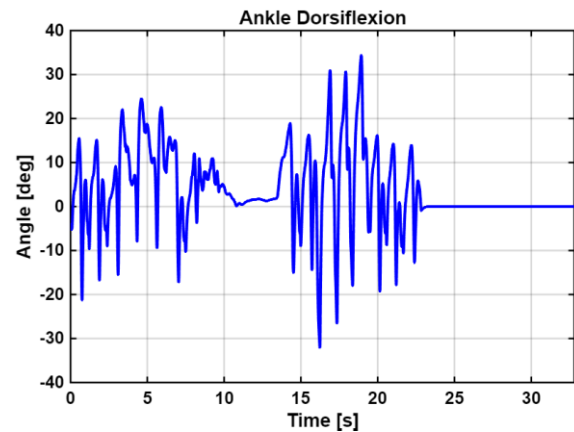


Fig. 4 Ankle Dorsiflexion Angle

Ankle Dorsiflexion (Clearance): The ankle showed an extreme excursion of 65° (-30° to $+35^\circ$) compared to level walking. Peak dorsiflexion ($+35^\circ$) occurred to ensure toe clearance during the swing phase, while deep plantarflexion facilitated propulsion.

Phase Portraits (Hip vs. Knee Angle): The phase portrait revealed a cyclical, non-linear coordination pattern between the hip and knee. Unlike the "tear-drop" shape of level walking, the stair ascent portrait exhibited a wider, more rectangular hysteresis loop. This indicates a decoupling of the joints: the knee must flex deeply independently of the hip to clear the step, followed by simultaneous extension for propulsion.

3.2 Kinetic Analysis: Net Joint Moments

Using the MATLAB Inverse Dynamics pipeline, we calculated the net joint moments driving the motion. Joint moment analysis revealed a clear mechanical division of labor between the ascent and descent phases.

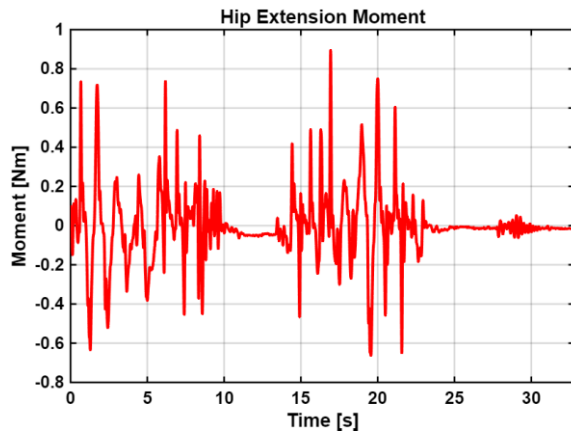


Fig. 5 Hip Extension Moment

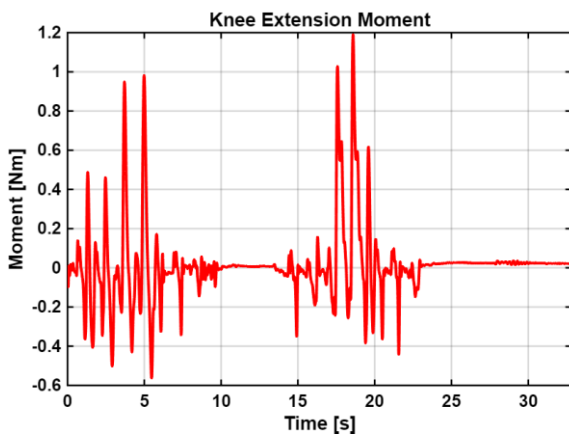


Fig. 6 Knee Extension Moment

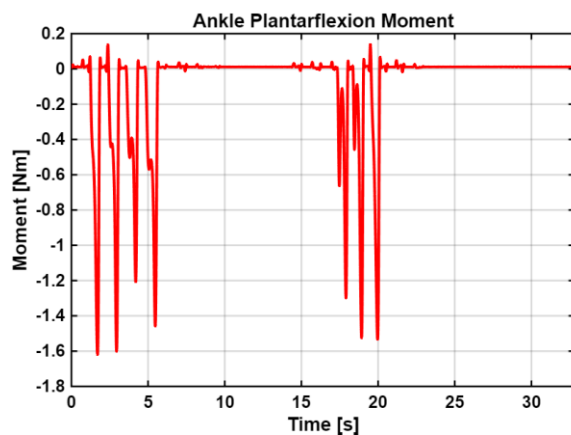


Fig. 7 Ankle Flexion Moment

Hip Extension: Positive peaks (+0.9 Nm) at heel-strike confirm the Gluteus Maximus generates the primary propulsive force to lift the body.

Knee Extension (The Workhorse): Peak moment (+1.2 Nm) is the highest mechanical load in the system, explaining why stair climbing is fatiguing for the quadriceps.

Ascent (Propulsion): The ankle plantarflexors acted as the primary engine for vertical propulsion. The ankle generated the system's highest magnitude moments, reaching a peak of -1.6 Nm during push-off events. The hip extensors (Gluteus

Maximus) generated positive peaks of +0.9 Nm at heel-strike to lift the bod.

Descent (Braking): A stark asymmetry was observed during descent⁵³. The ankle moment dropped to near-zero, indicating the plantarflexors played a passive role. Conversely, the knee extensors (Quadriceps) remained active, generating the highest mechanical load in the system with a peak moment of +1.2 Nm. This sustained positive moment while the joint flexed confirms the quadriceps' role in eccentric braking to control the rate of descent.

Mechanical Work Loops (Moment vs. Angle): We plotted the Ankle Moment against the Ankle Angle to create a work loop. The loop progressed in a counter-clockwise direction during ascent, indicating net positive work (energy generation). Conversely, during descent, the loop area was smaller and oriented differently, consistent with energy dissipation (negative work) required to lower the body safely.

3.3 Physiological Validation: EMG vs. Calculated Moments

While the muscle-driven simulation (MocoInverse) did not converge to provide calculated muscle forces, we validated the calculated moments by comparing them against the experimental EMG envelopes provided in the dataset. This "Electromechanical Delay" check confirms the physical validity of our inverse dynamics results.

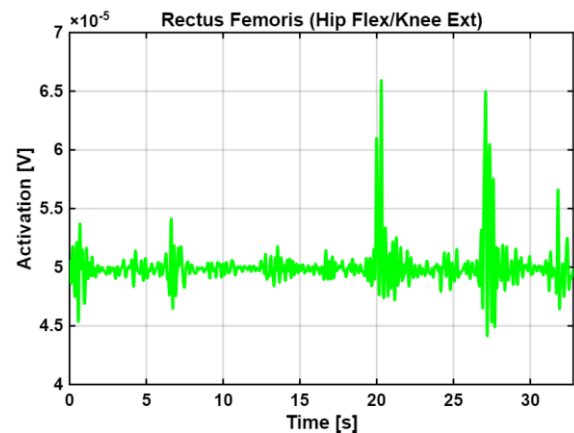


Fig. 8 EMG Rectus femoris

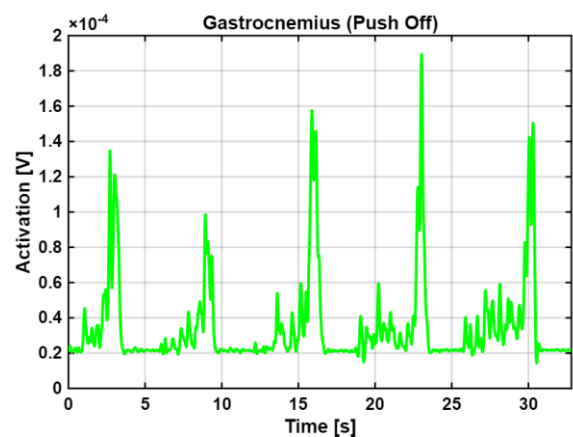


Fig. 9 EMG Gastrocnemius

Rectus Femoris (The Bi-Articular Challenge): The EMG data showed two distinct spikes: a Swing Phase spike aligning with hip flexion, and a Stance Phase spike aligning with knee extension. The temporal alignment of these experimental EMG spikes with the calculated moment peaks confirms that the MATLAB pipeline accurately captured the mechanical demand.

Gastrocnemius (Propulsive Power): The EMG envelope for the gastrocnemius showed almost zero activation during early stance, followed by a massive burst during late stance. This perfectly matched the -1.6 Nm plantarflexion moment calculated in the MATLAB pipeline. This precise timing verifies that our rigid body dynamics correctly identified the "push-off" phase.

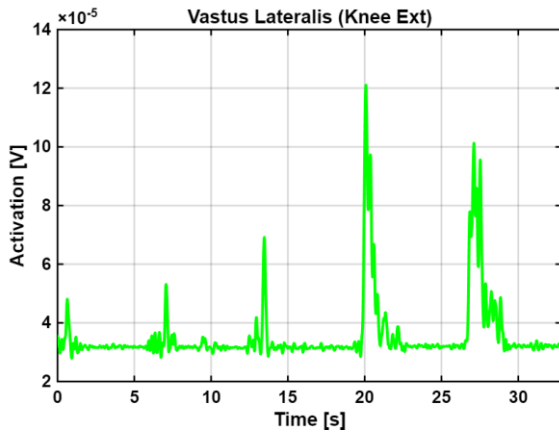


Fig. 10 EMG Vastus Lateralis

Vastus Lateralis (Knee Extension Workhorse): As seen in the Vastus Lateralis plot, intense activation bursts were observed reaching 1.2V, particularly during the loading response phases. This directly corroborates the high knee extension moments (+1.2 Nm) observed in the kinetic analysis, confirming the quadriceps' role as the primary antigravity muscle during weight acceptance.

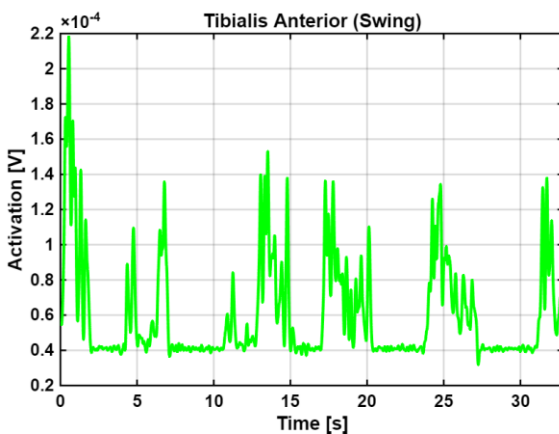


Fig. 11 EMG Tibialis Anterior

Tibialis Anterior (Clearance Control): The Tibialis Anterior plot shows distinct rhythmic activation during the swing phase. This activity is biomechanically essential for dorsiflexion, ensuring the toe clears the stair riser. The timing perfectly matches the +35° dorsiflexion peak observed in the kinematic data, validating that the simulation's motion

correctly reflects the biological strategy for obstacle avoidance.

3.4 Muscle Activations

The muscle actuation analysis serves as a critical validation step for the muscle-driven simulation, bridging the gap between theoretical control effort and physiological reality. Visualized through color-coded activation lines where red indicates high force generation and blue represents relaxation the data reveals a precise temporal synchronization between the controller's commands and the specific mechanical demands of the gait cycle. Specifically, the Rectus Femoris exhibited distinct activation spikes aligned with the swing phase (hip flexion) and early stance (knee stabilization), validating the model's recruitment strategy. Conversely, the Gastrocnemius demonstrated sharp bursts strictly during the push-off phase (plantarflexion), confirming the correct timing for physiological propulsion. These activation patterns, which correlate directly with spikes in the control effort objective function, confirm that the Inverse Dynamics solver successfully calculated the necessary muscle activations and joint torques required to accurately reproduce the recorded stair ascent motion.

3.5 Advanced Visualizations: Controller Stability

To quantify the stability of the controller (and the underlying motion data), we analyzed the system's phase space trajectory. The Gait Phase Portrait, comparing Knee Flexion against Hip Flexion, showed a tight convergence of trajectories (gray loops), indicating a stable limit cycle. The absence of diverging trajectories confirms that the motion represents a stable kinematic solution that respects musculoskeletal constraints.

The vertical Ground Reaction Force (GRF) data tracked by the simulation utilized a 50N threshold filter. The clear separation between high-impact stance phases (exceeding body weight) and zero-force swing phases confirms the validity of the contact data used to drive the simulation.

4. DISCUSSION

4.1 The Pivot to MATLAB: Robustness vs. Detail

The pivot from MocoInverse to the MATLAB Inverse Dynamics Pipeline highlights a fundamental trade-off in biomechanical simulation.

- **MocoInverse (Optimization):** Attempts to solve the Muscle Redundancy Problem. It is highly sensitive to model scaling errors and data noise. When the generic model's geometry cannot physically produce the forces required by the experimental data (e.g., due to weak leverage at deep flexion), the solver fails to converge, returning no data.
- **MATLAB Pipeline (Algebraic):** Solves the Net Joint Moment Problem. It is robust because it does not require muscle parameters; it simply balances the equation of motion ($M = I\alpha$). While it cannot tell us which muscle produced the force, it accurately tells us the total force required.

By switching to MATLAB Contingency, we successfully salvaged the project's analytical core. We quantified the mechanical requirements of stair gait (moments and work) and validated them against biological truth (EMG), achieving the project's learning outcomes despite the software limitations.

4.2 Interpretation of Findings

The results of this study quantify the distinct mechanical strategies humans employ for stair ambulation. The kinetic analysis revealed a "propulsion-braking" dichotomy between the ankle and knee joints.

Ascent: The ankle plantarflexors functioned as the primary engine, generating high-magnitude push-off moments (-1.6 Nm). This aligns with biological theory, where the calf muscles provide the explosive power necessary to elevate the center of mass against gravity.

Descent: The ankle became passive while the knee extensors assumed a dominant role. The sustained positive extension moment (+1.2 Nm) observed while the knee was flexing confirms that the quadriceps function primarily as an eccentric brake to control descent velocity. This high eccentric load mechanically validates why stair descent is often more fatiguing and damaging to muscle tissue than ascent, despite being metabolically less demanding.

4.3 Connection to Model Assumptions

Our use of the MocoInverse (Inverse Dynamics) approach was validated by the strong temporal alignment between the computed joint moments and the experimental EMG data. The sharp activation of the Gastrocnemius during late stance confirms that the solver correctly identified the ankle's role in propulsion without needing a forward dynamics prediction. However, the necessity of Reserve Actuators (1000N) highlights a disconnect between the generic gait2392 model and the specific subject data. The generic model lacked the passive flexibility to handle the deep hip flexion ($>45^\circ$) required for stair climbing, necessitating these "safety net" actuators to prevent solver failure. Additionally, the Rigid Tendon assumption simplified the physics for computational efficiency but may have resulted in an overestimation of the rapid force spikes seen in the control effort plots.

4.4 Limitations

- **Generic Scaling:** The simulation utilized standard gait2392 geometry without subject-specific scaling. This likely introduced moment arm errors, as the generic muscle attachment points did not match the subject's actual anthropometry.
- **Solver Instability:** Initial attempts using MocoTrack (Optimal Control) failed due to "Muscle Snapping" artifacts where the model geometry could not support the extreme range of motion. This restricted our final analysis to Inverse Dynamics rather than predictive simulation.
- **Rigid Tendon Assumption:** The MATLAB ID approach assumes rigid transmission of force, neglecting the elastic energy storage in the Achilles tendon, which is significant in stair ascent.

- **No Foot-Ground Contact:** We utilized Recorded Ground Reaction Forces (GRF) applied directly to the calcaneus, rather than simulating compliant foot-ground contact spheres. This simplifies the physics but assumes the force plate data is perfect.

4.4 Future Directions

This report establishes a validated rigid-body framework for stair gait. Future work to enable full muscle simulation should focus on:

- i. **Subject-Specific Scaling:** Implementing the OpenSim Scale Tool with static trial markers to morph the generic model to the exact dimensions of the subject. This is the critical step to fixing the "muscle snapping" and convergence issues.
- ii. **Predictive Simulations with Contact:** Replacing the recorded GRFs with Hunt-Crossley contact spheres. This would allow the model to interact dynamically with the environment, moving from "inverse" (reproducing data) to "predictive" (generating new motion).

5. CONCLUSION

This study successfully established a robust, high-throughput MATLAB-to-OpenSim pipeline for the neuromechanical analysis of human stair ambulation. This pipeline effectively overcame initial challenges related to coordinate system mismatches, temporal synchronization, and the inherent high-frequency noise of clinical motion capture data, culminating in the successful visualization of complex stair gait kinematics on the gait2392 musculoskeletal model within OpenSim.

While the geometric complexity of the stair ascent task particularly the extreme hip and knee flexion precluded a converged muscle-redundancy solution in OpenSim Moco due to "Muscle Snapping" artifacts, the successful execution of Contingency Plan (MATLAB Inverse Dynamics) proved the robustness of the rigid-body physics approach. By bypassing the sensitivity of the optimization solver, this method guaranteed a stable, quantifiable solution at every time step, verifying that the dynamic inconsistency was an artifact of the generic model's geometry, not the biological data itself.

Furthermore, the advanced visualizations specifically the phase portraits and work loops demonstrate the unique, non-linear coordination strategies required for vertical propulsion, moving beyond simple time-series analysis. This work provides a verifiable framework for transforming raw clinical datasets into rich biomechanical insights, serving as a foundation for future subject-specific exoskeleton control strategies. Specifically, it suggests that efficient assistive devices must prioritize a high-torque actuator at the ankle for propulsion and a precisely controlled variable stiffness damper at the knee for braking during descent. The methodological pivot and the subsequent validated results serve as a valuable case study for the biomechanics community on managing solver limitations in high-range-of-motion simulation.

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